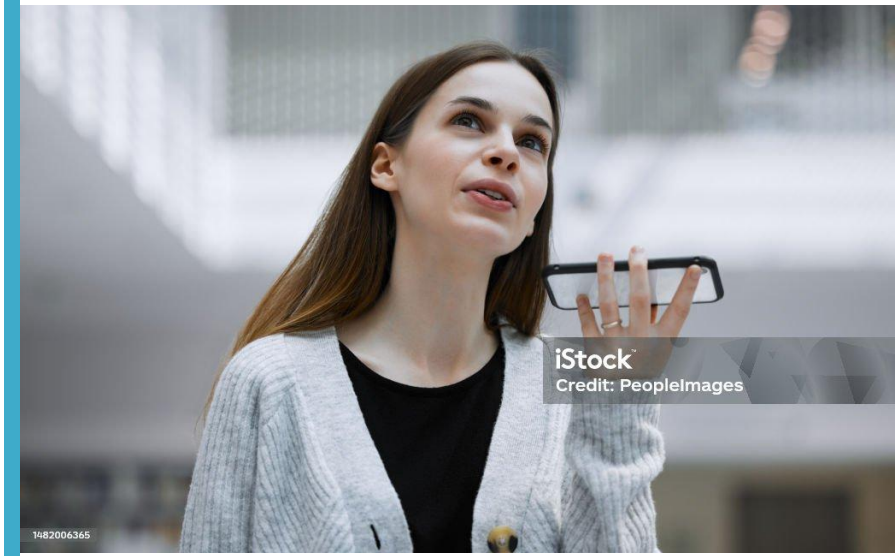
Introduction to AI & Generative AI

Artificial Intelligence, or AI, is a way of making computers think and learn like humans.

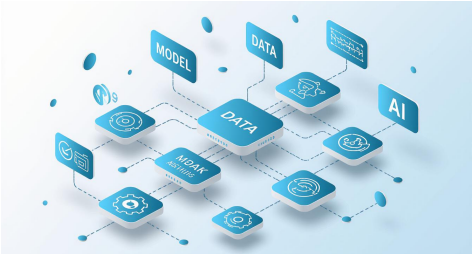
- It mimics human intelligence to solve problems and make decisions.
- Voice Assistants: Siri and Alexa understand your voice and answer questions.
- **Self-Driving Cars**: Cars use AI to "see" the road and navigate safely.
- It is not just about robots; it is the "brain" inside the software we use every day.



Evolution of AI

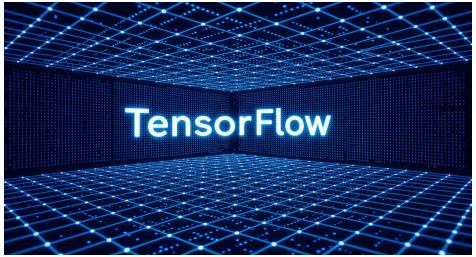
Artificial Intelligence

The broad concept of machines acting smartly.



Machine Learning

A subset where computers learn from patterns in data.



Deep Learning

Using "neural networks" to solve very complex problems.

What is Artificial Intelligence (AI)?

AI is all around us, helping us make better choices and saving us time.

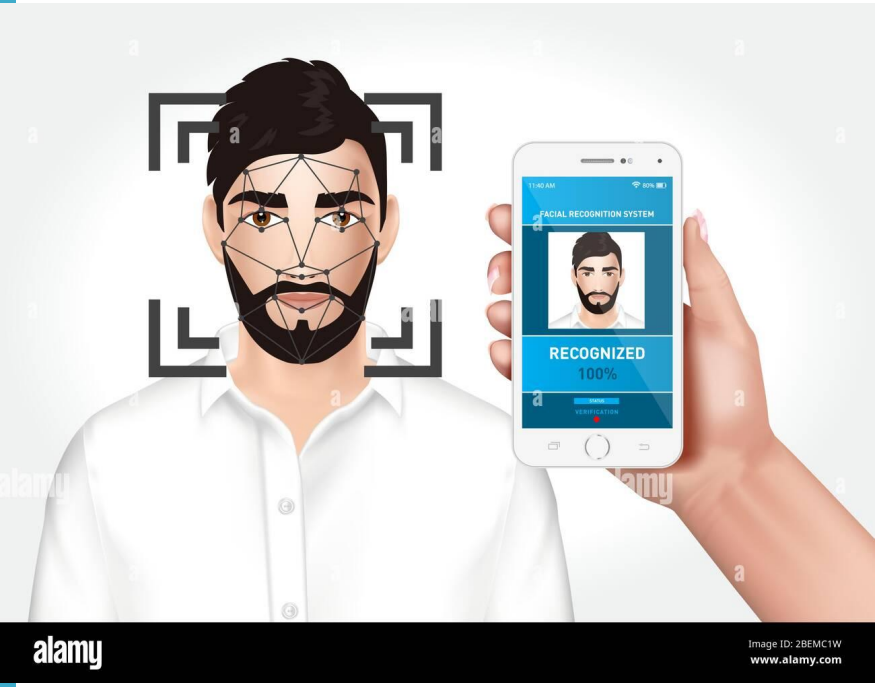
- Google Maps: Predicts traffic and finds the fastest route for you.
- Netflix Recommendations: Suggests movies based on what you liked before.
- Siri/Google Assistant: Understands your intent when you ask for a timer or a joke.
- It is like having a **digital assistant** that learns your habits.

What is Machine Learning (ML)?

Machine Learning is a type of **AI** that learns by looking at lots of **data** instead of following strict rules.

Think of it like a student practicing math problems to get better.

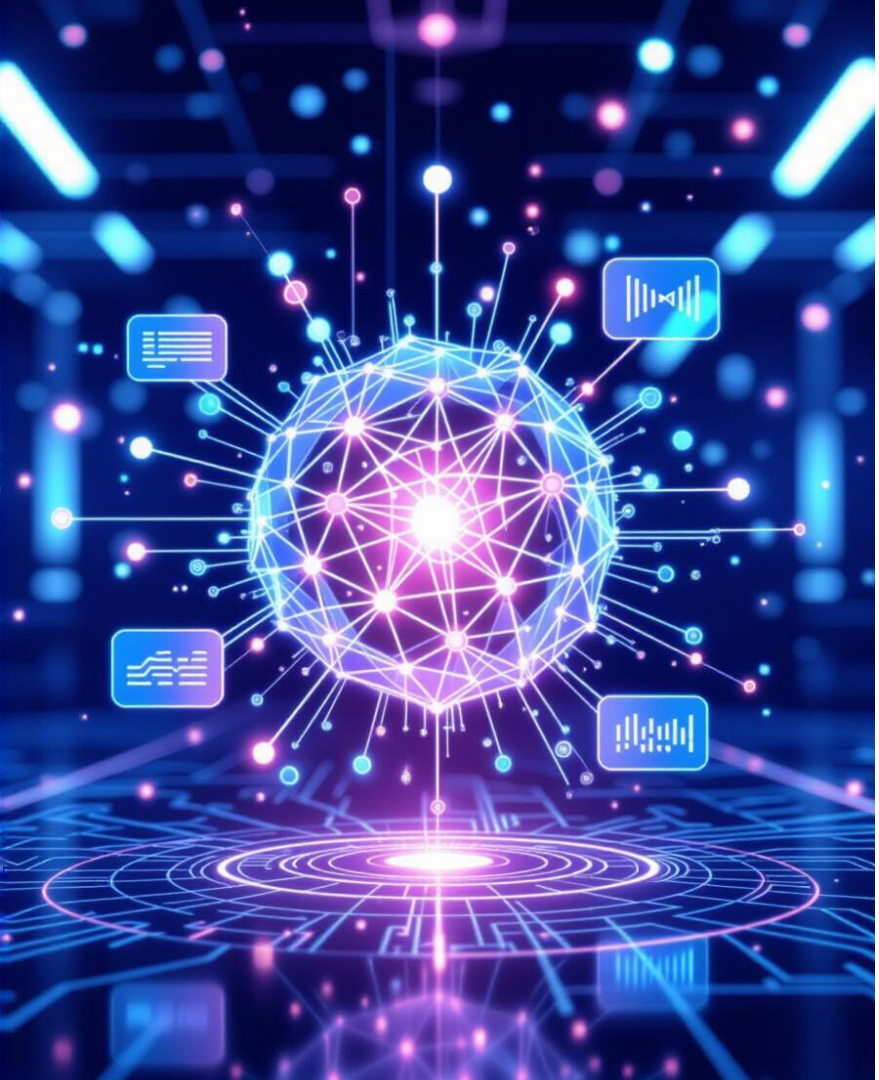
Fraud Detection: Banks use ML to spot unusual spending patterns. If you suddenly buy something in a different country, the ML model flags it as 'suspicious' because it doesn't match your usual data.



What is Deep Learning?

Deep Learning is a more advanced version of Machine Learning that uses "layers" to think.

- It is inspired by how the human brain works.
- **Face Recognition**: Your smartphone uses deep learning to identify your face among millions of others.
- It looks at tiny details like the distance between your eyes or the shape of your nose to be 100% sure it is you.



What is Generative AI?

Generative AI is a special type of AI that can create new content from scratch.

- Text: ChatGPT can write essays, poems, or computer code.
- Images: Tools like Midjourney or DALL-E can create art from a text description.
- Music: AI can now compose original songs in the style of your favorite artist.
- It doesn't just find information; it **builds** something that didn't exist before.

Deep Learning vs Generative AI

While they are related, they have different "jobs":

- Deep Learning (The Predictor): It looks at data to make a guess.
  - Example: "Is this a picture of a cat or a dog?"
- Generative AI (The Creator): It uses what it knows to make something new.
  - Example: "Draw me a picture of a cat wearing a space suit."
- One identifies, the other **invents**.

What is a Neural Network?

A **Neural Network** is the 'engine' behind modern **AI**.

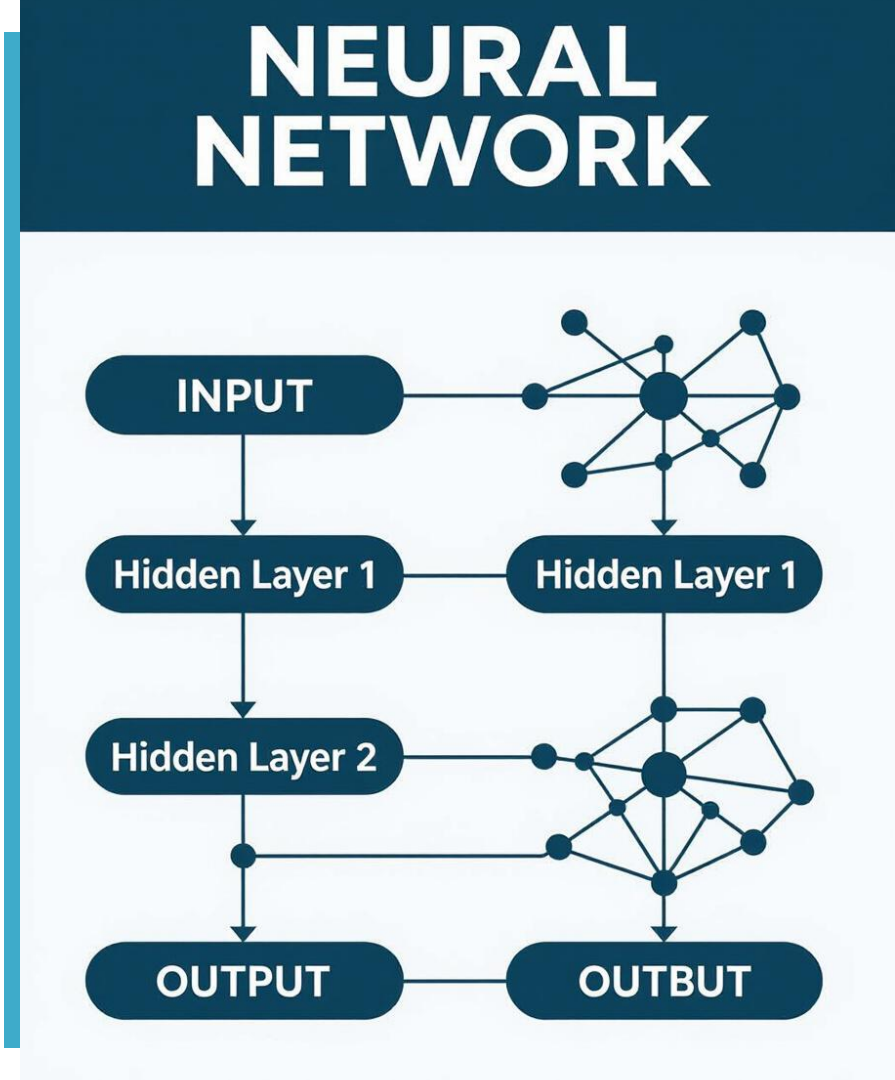
It is made of tiny units called **neurons**, just like your brain.

These digital neurons pass information to each other. When a neuron gets a piece of info, it decides if it is important enough to pass to the next one. By working together, millions of neurons can solve **massive problems**.

Structure of Neural Networks

Neural networks are organized into three main parts:

- Input Layer: This is where the data enters (like the pixels of a photo).
- Hidden Layers: This is where the "thinking" happens. There can be many layers here.
- Output Layer: This is the final answer (like saying "That is a cat!").
- **Analogy**: It is like a factory line where each station adds a bit more detailto the product.



Activation Function

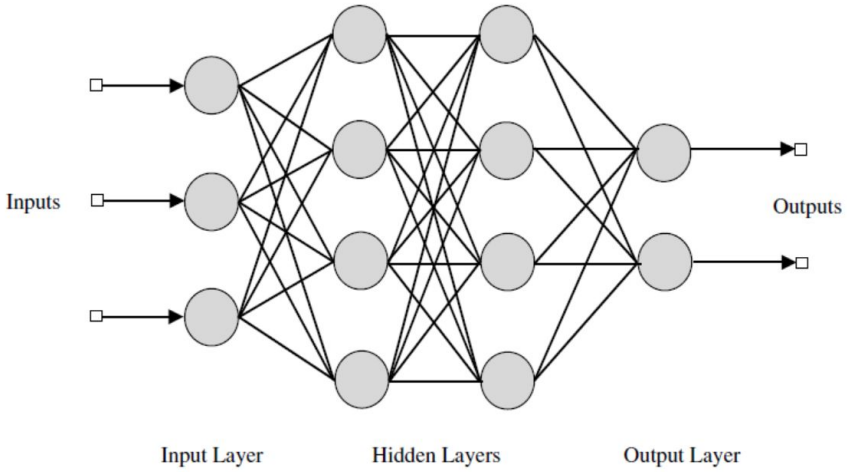
The Activation Function is like a gatekeeper for information.

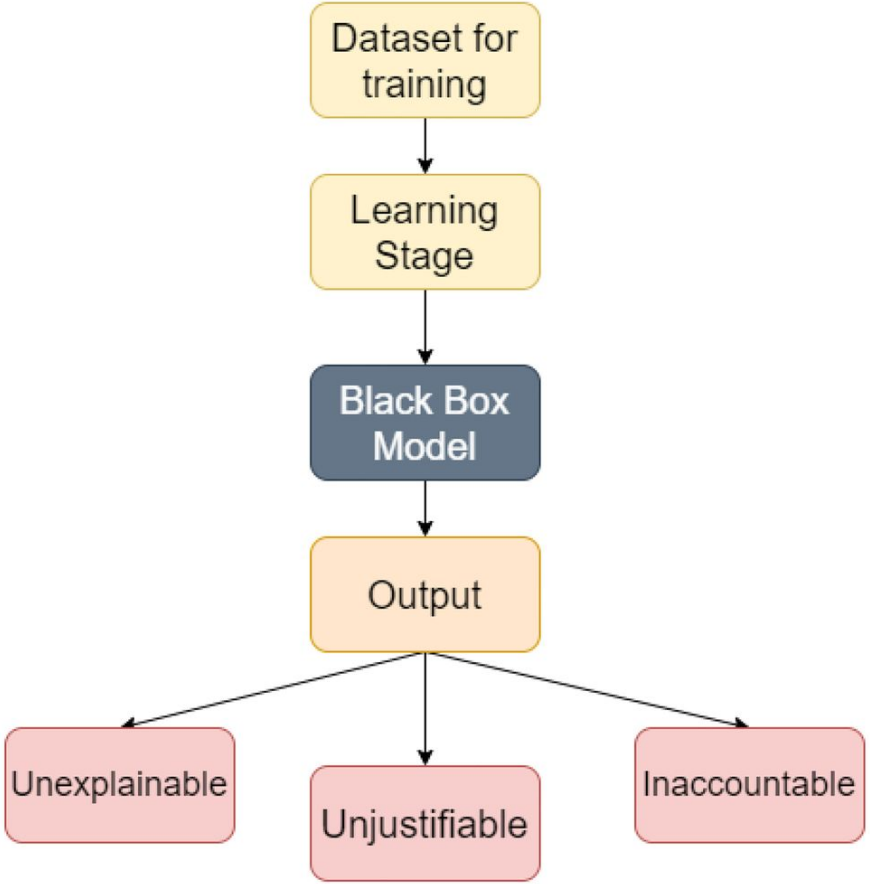
- It decides whether a neuron should "fire" (pass info) or stay silent.
- Why it matters: Real life isn't a straight line; it is complex and curvy.
- **Example**: If you are deciding to go outside, you only go if the "Sunlight" signal is strong enough. The activation function handles that "Yes/No" logic.

Types of Neural Networks

Different tasks require different types of "brains":

- Feedforward: The simplest type; info moves in one direction.
- CNN (Convolutional): Best for images. It "scans" pictures to find edges and shapes.
- RNN (Recurrent): Best for **sequences** like speech or music because it remembers what happened just a second ago.





Generative AI Model (Black Box)

Think of a Generative AI model as a "Black Box" that takes a request and gives a result.

- Prompt: This is your instruction (e.g., "Write a story about a brave toaster").
- Completion: This is the AI's response or the finished story.
- The "Black Box" uses its massive training to turn your words into a **uniquecreation**.

Real-time Example of AI

We see AI in action every single day:

- Chatbots: Customer service bots that help you track a package instantly.
- Image Generation: Creating a logo for a new business just by describing it.
- Language Translation: Apps that translate signs in real-time using your camera.
- These tools save hours of manual work by doing it in **seconds**.

Foundation Models

Foundation models are the 'super-brains' of the **AI** world.

They are trained on **huge amounts of data** (like the whole internet!).

Instead of being good at just one thing, they are **adaptable**. One foundation model can help write a book, summarize a meeting, or even help a scientist discover a new medicine.

Large Language Models (LLMs)

Learning Patterns

They learn the patterns of how we speak and write.

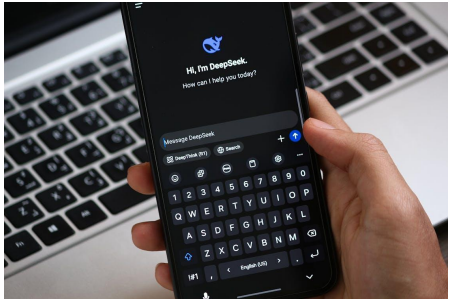
Prediction Accuracy

They predict the **next most likely word** in a sentence.

Human-like Predictions

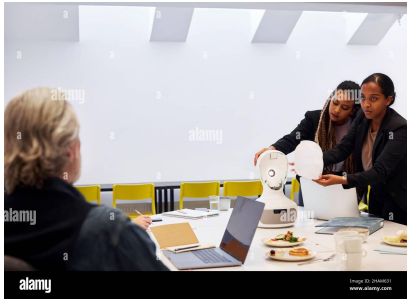
Because they have read so much, their predictions are incredibly accurate and sound very human.

Natural Language Processing (NLP)



Bridging Computer Code and Human Talk

Natural Language Processing (NLP) is a field of AI that enables computers to **read** and **understand** human language. It serves as a bridge between **computer code** and **human talk**, allowing machines to interpret the nuances, slang, and rhythm of human speech.



Enhancing Communication with Machines

Without NLP, interacting with machines would require users to input commands in a programming language. NLP empowers machines to comprehend human language, enhancing communication by enabling them to understand context and nuances in speech.

NLP Process

How does a machine handle a sentence? It follows a simple path:

- Input: You say or type something (e.g., "What's the weather?")
- Understanding: The AI breaks the sentence down to find the meaning.
- Response Generation: The AI looks up the data and builds a natural-sounding answer.
- It happens so fast it feels like a **real conversation**.

Natural Language Understanding (NLU)

NLU is the "listening" part of the process. It focuses on intent:

- If you say "It's chilly in here," NLU helps the AI understand you want the heat turned up.
- It looks past the literal words to find the hidden meaning.
- **Example**: Understanding that "I'm feeling blue" means you are sad, not that you changed color.

Natural Language Generation (NLG)

NLG is the "speaking" part of the process. It focuses on output.

- It takes raw data and turns it into a sentence a human would say.
- Instead of saying "Temperature 72, Sky Clear," NLG says, "It's a beautiful sunny day with a high of 72 degrees."
- It makes AI feel **friendly and helpful** rather than robotic.

How LLMs Work

LLMs follow a three-step process to talk to you:

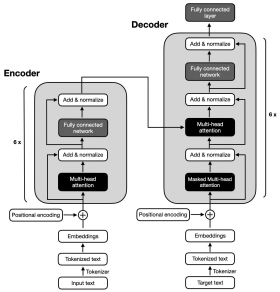
- Tokenization: Breaking your sentence into small chunks called "tokens" (words or parts of words).
- Understanding: Analyzing the relationship between all those tokens.
- Response: Picking the best tokens to build an answer, one by one.
- It is like a very fast game of **word association**.

Transformer Architecture

The "Transformer" is the specific technology that made ChatGPT possible.

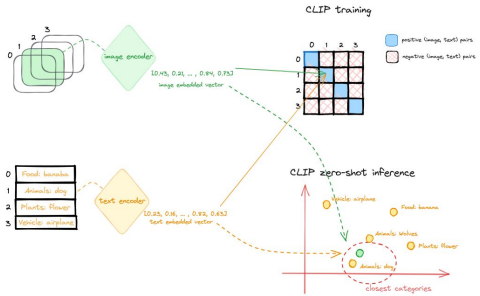
- Encoder: Reads and understands the input you provide.
- Decoder: Takes that understanding and starts "decoding" it into a response.
- It allows the AI to focus on the **most important words** in a long sentence, which is called "Attention."

Encoder vs Decoder



Encoder: The Reader

The **Encoder** in AI is proficient at understanding context and is commonly used in tasks such as classifying emails as "Spam" or "Not Spam." It focuses on interpreting input data and extracting meaningful information for further processing.



Decoder: The Writer

On the other hand, the **Decoder** in AI excels at generating new text based on the input it receives. It is utilized in tasks like generating responses, writing stories, or answering questions by producing coherent and relevant output.

Real-world Use Cases

Chatbots

Providing 24/7 support for websites.

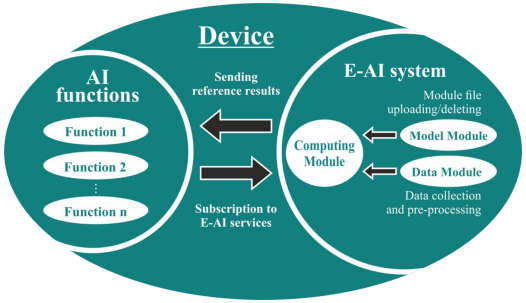
Content Creation

Helping writers overcome "writer's block" by suggesting ideas.

Recommendations

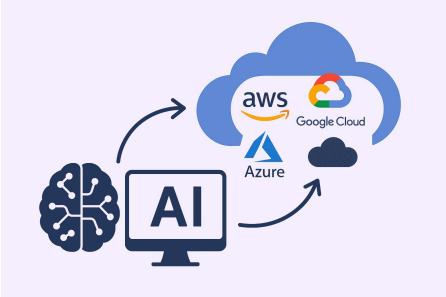
Helping you find your next favorite song on Spotify.

How AI Applications are Built



Embedded Models: Integration within Software

Developers integrate AI directly into applications through **Embedded Models**, where the AI functionalities are embedded within the software or device itself. This approach is commonly seen in applications like image recognition in smartphone cameras.



Hosted Models: Cloud-Based AI

Alternatively, AI can be deployed on powerful servers in the cloud, known as **Hosted Models**. In this setup, the application communicates with the AI model over the internet, leveraging the computational resources of the cloud for complex AI tasks.

Open vs Closed Source Models

There are two philosophies in the AI world:

Open Source: The "recipe" for the AI is public. Anyone can see it, fix it, or use it (Example: Llama by Meta).

Closed Source: The "recipe" is a secret owned by a company (Example: GPT-4 by OpenAI).

Open source encourages collaboration, while closed source often offers high security.

Tools and Platforms

You don't have to be a genius to start using AI tools:

- Hugging Face: Think of it as the "App Store" for AI models. You can find thousands of free models there.
- APIs: A "plug-in" that lets a regular app talk to a powerful AI like ChatGPT.
- These tools make it easy for anyone to **build their own AI app**.

Key Takeaways

Let's recap what we learned today:

- AI mimics human intelligence; Generative AI creates new things.
- Neural Networks are the brains, and Transformers are the modern architecture.
- NLP helps machines understand and speak our language.
- AI is a tool meant to **assist and enhance** what humans can do.

Conclusion

The future of AI is being written right now!

- AI will continue to get smarter, faster, and more helpful.
- It is a powerful tool for creativity and problem-solving.
- As students and beginners, learning how these "digital brains" work is your first step into the **future of technology**.

**Thank you! Any questions?